

Solution Design

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Project Name: AI-Powered Knowledge Assistant

Team Name: Hello World

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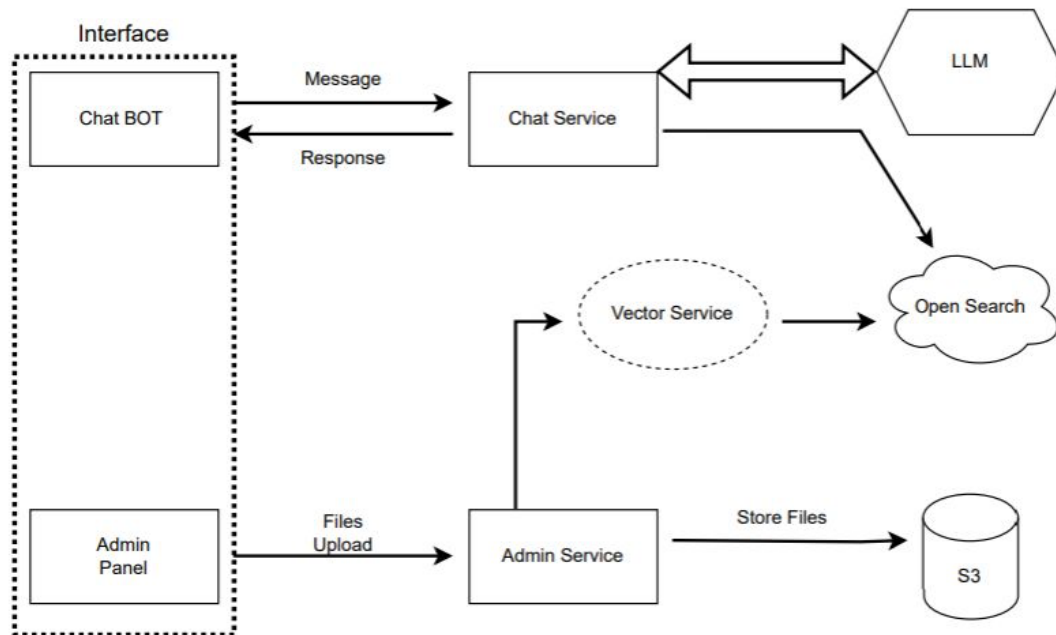
Overview :

The AI-Powered Knowledge Assistant enables users to upload documents (PDF/text) and engage in natural language conversations to retrieve information from them. The assistant supports real-time Q&A by parsing documents, storing content as vectors in OpenSearch, and using a powerful LLM (Qwen 3) to generate contextual responses.

Business Flow :

1. Document Upload (via Admin Panel)
2. Preprocessing & Embedding
3. Storage (S3 for documents, OpenSearch for vectors)
4. End-user Interaction (via Chat Bot)
5. Contextual Retrieval + LLM Response

Solution Design



Problem Statement

- **Primary Challenge:** Enable users to upload documents (PDF/Text), extract and store knowledge from them, and ask contextual questions with accurate, source-backed answers.
- **Target Audience:** Knowledge workers, support agents, internal teams, enterprises needing document-based Q&A systems.
- **Current State:** Manual document search is time-consuming; existing tools lack real-time semantic comprehension or self-hosted control.

Business Impact & Value Proposition

- **Revenue Impact:** Potential to monetize as a SaaS tool for document-based AI assistants
- **Cost Savings:** Saves 100+ hours/month in manual research, lowering support costs by 30–50%.
- **Customer Benefits:** Quick access to critical knowledge, 24x7 self-service capability, centralized document intelligence.
- **Innovation & Uniqueness:** Combines Nomic Embeddings + Qwen3 to provide semantic Q&A without relying on OpenAI, the Chat bot can be integrated into the partners website.

Solution Overview

- **Core Solution:** An AI assistant Chat Bot that parses user-uploaded documents, vectorizes their content, stores the vectors in OpenSearch, and retrieves relevant passages to answer user questions.
- **Key Features:**
 - Document Upload and Parsing through an Admin Panel
 - Sentence-level Vector Embedding (Nomic)
 - Semantic Search via KNN in OpenSearch
 - LLM-powered Answer Generation
 - React-based iframe embeddable Chat UI
- **Success Metrics:**
 - 90%+ answer relevance score
 - < 5s response time
 - 10K+ docs indexed without degradation
- **Implementation Scope:**
 - Upload, parse, vectorize, store, and query PDFs/Text
 - Embed via React iframe and MS Teams
 - Support OpenSearch + AWS S3 backend

Requirements

- **Functional Requirements:**
 - File upload
 - PDF parsing
 - Sentence chunking and vectorization
 - Vector storage in OpenSearch
 - LLM-based Q&A
 - Chat UI
- **Non-Functional Requirements:**
 - Response time < 5s
 - Horizontal scalability
 - Secure file handling and API communication
- **Business Requirements:**
 - Enable knowledge reuse
 - Lower manual research cost
 - Plug into existing workflows (Websites, Teams, Slack)
- **User Requirements:**
 - Upload docs easily
 - Ask any question
 - Get short, relevant, and source-anchored answers